

Snow cover estimation from MODIS and Sentinel-1 SAR data using machine learning algorithms in the western part of the Tianshan Mountains

LIU Yang^{1,2,3,4}  <https://orcid.org/0000-0001-5751-0982>; e-mail: liuyang_dsslab@163.com

CHEN Xi³  <https://orcid.org/0000-0003-0432-0733>; e-mail: chenxi@ms.xjb.ac.cn

HAO Jian-Sheng^{1,2,3,4}  <https://orcid.org/0000-0003-3296-5838>; e-mail: haojiansheng14@mails.ucas.ac.cn

LI Lan-hai^{1,2,3,4*}  <https://orcid.org/0000-0002-1914-0180>;  e-mail: lilh@ms.xjb.ac.cn

*Corresponding author

¹ State Key Laboratory of Desert and Oasis Ecology, Xinjiang Institute of Ecology and Geography, Chinese Academy of Sciences, Urumqi 830011, China

² Ili Station for Watershed Ecosystem Research, Chinese Academy of Sciences, Xinyuan 835800, China

³ Research Center for Ecology and Environment of Central Asia, Chinese Academy of Sciences, Urumqi 830011, China

⁴ Xinjiang Regional Center of Resources and Environmental Science Instrument, Chinese Academy of Sciences, Urumqi 830011, China

Citation: Liu Y, Chen X, Hao JS, et al. (2020) Snow cover estimation from MODIS and Sentinel-1 SAR data using machine learning algorithms in the western part of the Tianshan Mountains. *Journal of Mountain Science* 17(4). <https://doi.org/10.1007/s11629-019-5723-1>

© Science Press, Institute of Mountain Hazards and Environment, CAS and Springer-Verlag GmbH Germany, part of Springer Nature 2020

Abstract: Obtaining the spatial distribution of snow cover in mountainous areas using the optical image of remote sensing technology is difficult because of cloud and fog. In this study, the object-based principle component analysis–support vector machine (PCA–SVM) method is proposed for snow cover mapping through the integration of moderate-resolution imaging spectroradiometer (MODIS) snow cover products and the Sentinel-1 synthetic aperture radar (SAR) scattering characteristics. First, derived from the Sentinel-1A SAR images, the feature parameters, including VV/VH backscatter, scattering entropy, and scattering alpha, were used to describe the variations of snow and non-snow covers. Second, the optimum feature combinations of snow cover were formed from the feature parameters using the principle component analysis (PCA) algorithm.

Finally, using the optimum feature combinations, a snow cover map with a 20 m spatial resolution was extracted by means of an object-based SVM classifier. This method was applied in the study area of the Xinyuan County, which is located in the western part of the Tianshan Mountains in Xinjiang, China. The accuracies in this method were analyzed according to the data observed at different experimental sites. Results showed that the snow cover pixels of the extraction were less than those in the actual situation ($FB_1=93.86$, $FB_2=59.78$). The evaluation of the threat score (TS), probability of detection (POD), and false alarm ratio (FAR) for the snow-covered pixels obtained from the two-stage SAR images were different ($TS_1=86.84$, $POD_1=90.10$, $FAR_1=4.01$; $TS_2=56.40$, $POD_2=57.62$, $FAR_2=3.62$). False and misclassifications of the snow cover and non-snow cover pixels were found. Although the classifications were not highly accurate, the approach showed potential for integrating different sources to retrieve

Received: 08-Aug-2019

Revised: 23-Oct-2019

Accepted: 23-Dec-2019

the spatial distribution of snow covers during a stable period.

Keywords: Snow cover; Estimation; Sentinel-1/2; MODIS; Machine learning

Introduction

The snowpack is one of the most active natural components of the cryosphere (Gascoin et al. 2019). Its broad geographic distribution with special physical features (e.g., high emissivity, high albedo, large melting latent heat, and low thermal conductivity) significantly affects the surface energy budget and climate change (Zhang 2005; Liu et al. 2015). Therefore, the variations of the snowpack are important in the energy exchange among land, atmosphere, and water, which generates direct and indirect effects on the utilization and management of water resources, including farming, forestry, and animal husbandry (Zimov et al. 2006; Tamocai et al. 2009; Qiao and Wang 2019; Wu 2019; Iorman et al. 2019; Shui et al. 2019). Consequently, the timely evaluation of the regional snow cover distribution is important in hydrological applications and disaster prevention and mitigation (Marshall et al. 1994; Hao et al. 2018; Wei et al. 2017).

Since the 1960s, the satellite-borne sensor has been an important exploration tool for monitoring snow distribution on different spatial and temporal scales (Dietz et al. 2012; Liu et al. 2017). The current remote sensing technologies for snow monitoring include synthetic aperture radar (SAR), light detection and ranging, and passive microwave technologies (Hou et al. 2019; Liu et al. 2018; Kostadinov et al. 2019). According to the current remote sensing data, the most widely used electromagnetic spectra for monitoring snow cover are mainly in the visible and short-wave infrared bands (Berman et al. 2018; Kuter et al. 2018). If the study area is a flat terrain under cloudless conditions, the currently released snow cover products, such as MOD10A1/MYD10A1, can accurately reflect the variations (Zhang et al. 2019; Mityóityg et al. 2). However, topographic effects in mountainous areas, such as geometric distortion, radiation brightness change, and the local incidence angle effect, along with thick clouds and

harsh weather conditions can seriously affect the accuracy of optical remote sensing data (Tsai et al. 2019). Therefore, information on the distribution of snow cover in such circumstances is not available (Masson et al. 2018). Researchers have attempted to remove cloud and its shadow from snow products using multi-temporal data fusion, spatial data fusion, and multi-source data fusion methods (Li et al. 2019; Parajka and Blöschl 2006; Masson et al. 2018). The multi-temporal data fusion of moderate-resolution imaging spectroradiometer (MODIS) snow products with Terra and Aqua sensors on a certain time scale has been found to reduce cloud coverage by 10%–20% (Gafurov et al. 2009; Li et al. 2019). In using spatial data fusion methods, topographic factors and interannual variation characteristics of snow cover are introduced to completely remove the cloud of MODIS snow cover (Li et al. 2017). Further, the cloud removal method of multi-source data fusion is extensively used by many researchers (Li et al. 2019). Currently, this method mainly uses AMSR-E snow water equivalent data to complement the cloud area of MODIS snow cover data. The disadvantage of this method is that, although it effectively reduces or removes a cloud-covered distribution by sacrificing the temporal or spatial resolution of MODIS snow products, it is difficult to extract the closed boundary of snow cover when the snow cover is fragmented or masked by massive clouds (Gao et al. 2010). Moreover, in a complicated terrain, the high heterogeneity of the local climate and the underlying surface easily lead to uncertainties in the regional snow cover estimation (Li et al. 2017; Li et al. 2019).

The distribution of snow cover can also be obtained using algorithms for supervised and unsupervised classifications (Forman and Reichle 2015; Xiao et al. 2018; Huang et al. 2018; Salzano et al. 2018). The application of an object-oriented classification in remote sensing is usually based on per-pixel classifiers (Xu et al. 2017). This technique assigns individual image pixels with a user-defined class based on the spectral characteristics of the individual pixels. Pixel-based classification is determined either computationally with minimum user input (unsupervised) or through user-defined training pixels (supervised) (Whyte et al. 2018). It realizes the high-precision classification of remote

sensing images through its precise targeting, high accuracy, and efficiency. It represents the trend of extraction for remote sensing information (Whyte et al. 2018; Lima et al. 2015). Through years of development, the support vector machine (SVM) algorithm has been widely used in the classification of remote sensing images. SVM based on a small sample and less experience is suitable for a nonlinear, high-dimensional pattern for processing the classification of remote sensing images. As an algorithm used for knowledge representation and reasoning, SVM has a huge potential in the application of a multi-source, heterogeneous remote sensing satellite for snow cover detection. Some researchers have been able to obtain the spatial distribution of snow cover successfully using SVM (Xiao et al. 2018; Nijhawan et al. 2019). However, few researchers have obtained the spatial distribution of snow cover by combining optical sensors and SAR sensors, which can detect through cloud and fog. Samples from snow cover products and SAR data sources are used in supervised classification, which exploits the respective advantages of optics and active microwave remote sensing. Moreover, the distribution of snow cover in the target area is obtained in a timely manner.

The Tianshan Mountains are located in Xinjiang, China, and they have a length of 2,500 km and width of 250–350 km from east to west. The area consists of large mountains, inter-mountain basins, and valleys. The terrain is complex with many variations, and the overall elevation decreases from west to east (Lu et al. 2016). The affluent snow mainly constitutes water resources in the arid region. Uncertainty in the spatial and temporal distributions of snow cover is intensified because of the dynamic changes in snow cover from the energy budget, which stems from different environmental factors, such as temperature and solar radiation in various elevations. Therefore, there is an urgent need to explore a scheme that can be used to obtain the spatial distribution of snow cover in complex terrain areas.

Based on the multi-source, heterogeneous stereoscopic observation data, including the MODIS snow cover product and Sentinel-1A SAR data, this study explores a scheme for snow cover detection using the combination of SAR and optical remote sensing data. This scheme overcomes the

interference imposed by cloud and fog from the optical sensing data and further expands active microwave sensing in snow cover detection. It has the advantages of machine learning and breaking through the limitation of a single remote sensing data source and a statistical algorithm to identify snow cover. Consequently, a high-resolution spatial distribution of snow cover in a complex mountainous terrain is obtained in a timely manner.

1 Data and Pre-processing

1.1 Study site

The target study area is located in Xinyuan County in the western part of the Tianshan Mountains, Xinjiang, China (Figure 1). Xinyuan County is known as the “frontier area with abundant water and grass” and “the wet island of Tianshan Mountains”. With a total area of 9,970 km², it is 191 km in length from east to west and 33.8 km in width from south to north. Xinyuan County stands higher in the east and lower in the west. It is surrounded by mountains on three sides with a west opening, forming a special terrain of a long east-to-west and a narrow north-to-south shape. It has a height of 820–3,813 m.a.s.l., with the lowest elevation of the western valley bottom of 820 m.a.s.l. and the highest peak of the eastern valley of 3813 m.a.s.l. Kunes River is the river at the bottom of the valley running from east to west, and they form special landforms with three mountain ranges in two valleys. This complex landform can be divided into the alpine

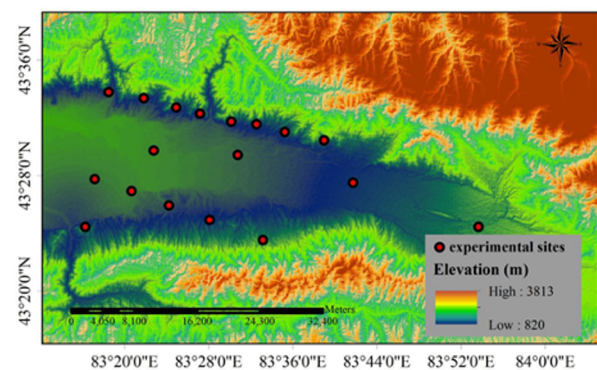


Figure 1 Sketch map of the study area in Xinyuan County, western part of the Tianshan Mountains.

Table 1 Remote sensing acquisition dates used in this study

Sentinel-1				Sentinel-2	MODIS
Activity image	Reference image	Geometry	Track	Acquisition image	Acquisition image
Jan. 29, 2018	Nov. 30, 2017	descending	165	Jan. 22, 2018	Feb.10, 2018
Feb.10, 2018	Nov. 30, 2017	descending	165	Feb 21, 2018	
Feb 22, 2018	Nov. 30, 2017	descending	165		

zone, middle mountain zone, low mountain zone, flatland, and hilly land. It can also be segregated into three geomorphic units from west to east: flat land, hilly land, and mountainous area.

Xinyuan County is a continental semi-arid climate zone with an average annual precipitation of 476 mm. The average annual temperature is 7.5 °C, and the average annual evaporation is 1285.8 mm. Its climate characteristics vary with the topography.

With high mountains serving as natural barriers in the valley, the cold and hot air flow from the southern and northern parts is blocked. The vegetation types are heterogeneous with a complete and complex vertical band spectrum through a vertical distribution. The main vegetation types are mountainous grassland and meadow. The semi-desert vegetation in the flat land is gradually being occupied by steppe and grassland vegetation.

1.2 Data collection and pre-processing

1.2.1 Test sites and data collection

The parameters of the snow characteristics in western Tianshan from 2017 to 2018 were provided by the Tianshan avalanche research team of the Xinjiang Institute of Ecology and Geography, Chinese Academy of Sciences. The samples of the study area are presented in [Figure 1](#). The observation parameters of the physical properties of snow included depth, temperature, humidity, density, type, and particle size of the snowpack. Each parameter was in accordance with the standard measurement method set by the International Association of Cryospheric Sciences.

Snow characteristics were collected during a stable period of the snowpack. Ground observation time was closely synchronized with the transit time of the Sentinel-1/2 satellite.

1.2.2 SAR data pre-processing

Proposed by the European Commission and the European Space Agency, Sentinel-1 is an

important part of the Global Monitoring for Environment and Security. It is made up of two satellites: Sentinel-1A and Sentinel-1B. Sentinel-1A was launched on April 3, 2014 and Sentinel-1B on April 25, 2016. The revisit period of a single satellite is 12 days. After the two satellites are networked, the revisit period is 6–12 days using a free data-sharing method ([Salzano et al. 2019](#)).

There are four sensor modes of Sentinel-1 in total: stripmap model, interferometric wide swath, extra-wide swath, and wave mode. Level 1 has two product types: single-look complex (SLC) and ground range detected. The data acquisition mode used in this study was interferometric wide (IW) SLC with a width of 250 km and a spatial resolution of 5 m × 20 m. The data used were the S1 TOPS-mode SLC of the IW mode ([Table 1](#)). All radar images were processed by the Sentinel application software (SNAP). The platform provides a series of tools and functions for the Sentinel-1 image processing and analysis. Free from the limitation of computer performance in the radar image processing, image processing was used to pre-process and geocode the whole image and extract the VV/VH backscatter. The specific steps are as follows:

- 1) SAR data orbit determination. The precise orbit is determined according to the orbit parameters of the product files.

- 2) Elimination of thermal noise.

- 3) Radiation calibration. Radiation calibration is determined by the sigma calibration coefficients in the product files.

- 4) Speckle noise removal. Speckle noise is removed through the refined Lee wave filtering. The filtering window is 5 × 5.

- 5) Topographic correction. Radar images are corrected according to the STRM data in the image covered area.

H -alpha (H - α) decomposition was conducted. To calculate the dual-polarized H - α decomposition, a 2×2 coherency matrix (T_{total}) was created using the complex data for every image pixel. U served as an orthonormal matrix. This matrix is an adaption

of the 3×3 coherency matrix, which is commonly applied to quad-polarized data and was first proposed by Cloude (2007) (Eq. 1).

$$T_{total} = \begin{pmatrix} T_{11} & T_{12} \\ T_{12} & T_{22} \end{pmatrix} = U \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix} U^H \quad (1)$$

$$= \lambda_1 u_1 u_1^H + \lambda_2 u_2 u_2^H$$

Therefore, a single complex covariance matrix (T_{dual}) can be expanded into the weighted sum of two simpler matrices, thus allowing for the pseudo-probabilities (P_i) to be defined by the sorted eigenvalues (λ). Given the eigenvectors and probabilities, the scattering entropy (H) and scattering alpha (α) values can be derived per pixel, as shown in Eqs. 2–3 (Whyte et al. 2018):

$$H = \sum_{i=1}^2 -p_i \log_2 P_i \text{ and } \alpha = \sum_{i=1}^2 P_i \cos^{-1}(|u_{1i}|) \quad (2)$$

Where

$$P_i = \lambda_i / \sum_{j=1}^2 \lambda_j, j = 1, 2 \quad (3)$$

In Eq. 2, the range of H is 0–1. The H parameter can be regarded as the output of a stochastic calculation within a resolution unit based on the scattering mechanism. In other words, H represents the probability of an effective scattering. When H approaches 0, the main types of associated land surfaces are isotopically pure and mediate an effective scattering. When H approaches 1, random scattering occurs. When H lies between 0 and 1, the system changes from an isotropic simple scattering to a completely random scattering (Xie et al. 2018). Therefore, this study aims to determine the variations of the scattering entropy and the scattering angle corresponding to the snow cover on the ground surface during a stable period.

1.2.3 Processing of Sentinel-2 for the snow cover

Launched as part of the European Commission's Copernicus program on June 23, 2015, Sentinel-2 was designed specifically to deliver a wealth of images. Sentinel-2 monitored the snow cover in the study area from the green and short-wave infrared band with a high resolution (10–60 m). The selected data were in accordance with time point of Sentinel-1 (Table 1).

Moreover, these images were less cloudy enough to serve as validation data in this study.

Two cloud-free Sentinel-2 level 1C images were downloaded from the scientific data hub of the ESA (<https://scihub.copernicus.eu/dhus/>). The Sentinel-2 L2A data were generated by the Sen2cor 2.5.5 plug-in, which is included in the SNAP software provided by the ESA. The spatial resolution of the Sentinel-2 data varied from band to band, and thus the nearest neighbor interpolation was used for resampling the spatial resolution to 20 m images in each band. Snow cover was obtained by Eqs. 4–5 through band calculation (Figure 2).

$$NDSI = \frac{\rho_{green} - \rho_{SWIR}}{\rho_{green} + \rho_{SWIR}} \quad (4)$$

where ρ_{green} (resp. ρ_{SWIR}) is the slope-corrected surface reflectance in the green band (i.e., SWIR at 1.6 μm). The NDSI indicates that only snow surfaces are very bright in visible light and very dark in the shortwave infrared light. Turbid water surfaces, such as lakes or rivers, may also have a high NDSI value. Thus, an additional criterion in the red reflectance was used to avoid false snow detection in these areas. A cloud-free pixel is classified as snow if the following condition is true (Gascoin et al. 2019):

$$NDSI > n_i \text{ and } \rho_{red} > r_i \quad (5)$$

where ρ_{red} is the slope-corrected surface reflectance in the red band. n_i and r_i are two parameter pairs. The detailed parameter settings and data processing were previously given in Gascoin (2019).

1.2.4 Processing of the MODIS snow cover products

The Collection-6 Terra/Aqua MODIS Level 3 of 500 m daily snow cover products (MOD10A1 and MYD10A1) is archived in the National Snow and Ice Data Center (Vignols et al. 2019; Hall et al. 2007). According to the global criteria for snow, land surfaces are regarded as snow cover when the NDSI is greater than 0.4, the near-infrared reflectance (band 2) is greater than 0.11, and the band 4 reflectance is greater than 0.10. Otherwise, they are regarded as non-snow cover (Tran et al. 2019).

The MOD10A1 (Table 1) was processed by mosaic geographic correction and resampled by the

product possessing software MRT. UTM44 projection was the projection coordinate system used, and nearest neighbor interpolation was the sampling method adopted. The vector boundary of the study area was used as the tailor to obtain the distribution of snow cover in the study area (Figure 3).

1.3 Machine learning algorithms

1.3.1 Principle component analysis (PCA)

Invented by Karl Pearson (1901), PCA is a popular statistical procedure for feature extraction. It can obtain a few uncorrelated components from high-dimensional original inputs by calculating the eigenvectors of the covariance matrix.

Given a set of m dimensional input vectors, $x_i = (x_i(1), x_i(2), x_i(3), \dots, x_i(m))^T$, where $i=1, 2, \dots, n$. PCA transforms X_i into a new y_i using the following formula:

$$y_i = U^T x_i \quad (6)$$

where U is the $m \times m$ orthogonal matrix. Its j th column u_j is the j th eigenvector of the sample covariance matrix (7):

$$C = \frac{1}{n} \sum_{i=1}^n x_i x_i^T \quad (7)$$

In other words, PCA solves the eigenvalue problem of Eq. 8:

$$\lambda_j u_j = C u_j, \quad j = 1, 2, \dots, m \quad (8)$$

where λ_j is one of the eigenvalues of C , and u_j is the corresponding eigenvector. Based on the estimated u_j , the components of y_i are calculated as the orthogonal transformation of X_i :

$$y_j(j) = u_j^T x_i, \quad j = 1, 2, \dots, m \quad (9)$$

The new components are called principal components (PCs). The selection of the number of PCs is in accordance with the principle that the contribution rate of variance should be more than 85%. These PCs can reflect most of the information of the original variables and contain

information that does not overlap (Chen et al. 2019).

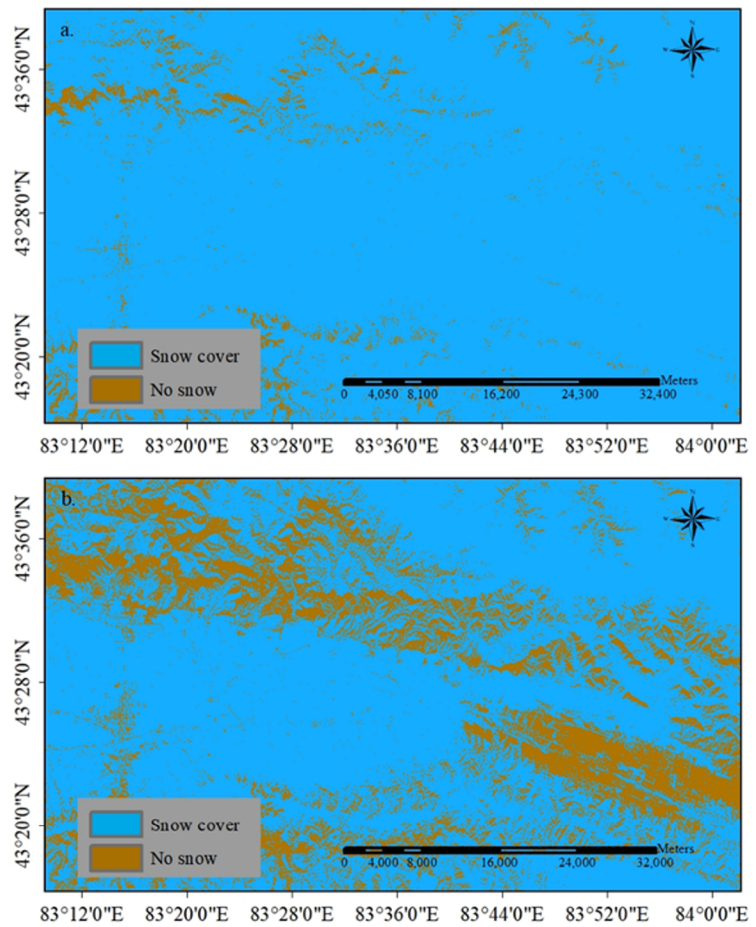


Figure 2 Spatial distribution of snow cover in Xinyuan County, western part of the Tianshan Mountains derived from the Sentinel-2 images on Jan. 22, 2018 (a), Feb 21, 2018 (b).

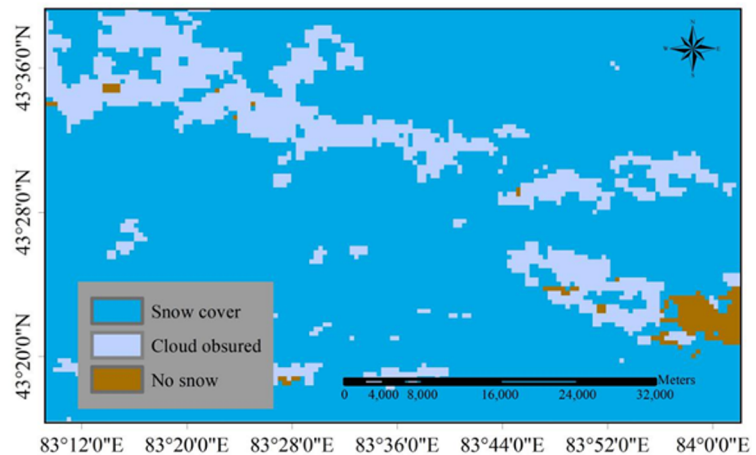


Figure 3 Map of snow cover distribution in Xinyuan County, western part of the Tianshan Mountains from the MODIS on Feb. 10, 2018.

1.3.2 Support Vector Machine (SVM)

SVM is a classical machine learning method developed on the basis of statistical theory. The idea of the SVM is to map the inseparable points in the low-dimensional space using a kernel function into linearly separable points in the high-dimensional space. The kernel function is the key to optimize the problems. To some extent, the kernel function guarantees the accuracy in selecting the appropriate kernel function in the SVM. The common kernel functions are linear kernel function, polynomial kernel function, radial basis kernel function, sigmoid kernel function, and Gaussian kernel function (Bangira et al. 2019).

1.4. Experimental Design

1.4.1 Feature generation for the classification

To examine the variations of the scattering characteristics of snow cover in contrast to those of non-snow cover, the following indicators served as the variables established from the VV/VH backscatter, scattering entropy, and scattering angle acquired from Sentinel-1 (Eqs. 10–16).

VH - scattering variations:

$$\Delta\sigma_{vh} = \sigma_{vh}^{avl} - \sigma_{vh}^{ref} \quad (10)$$

VV- scattering variations:

$$\Delta\sigma_{vv} = \sigma_{vv}^{avl} - \sigma_{vv}^{ref} \quad (11)$$

$\Delta\sigma_{vh/vv}$ - Cross-polarized ratio variations:

$$\Delta\sigma_{vh/vv} = \left(\frac{\sigma_{vh}^0}{\sigma_{vv}^0} \right)^{avl} - \left(\frac{\sigma_{vh}^0}{\sigma_{vv}^0} \right)^{ref} \quad (12)$$

$\Delta\sigma_{vv-vh}$ - Polarization difference variations:

$$\Delta\sigma_{vv-vh} = (\sigma_{vv}^0 - \sigma_{vh}^0)^{avl} - (\sigma_{vv}^0 - \sigma_{vh}^0)^{ref} \quad (13)$$

$\Delta\sigma_{(vv+vh)/2}$ - Polarization mean variations:

$$\Delta\sigma_{(vv+vh)/2} = \left(\frac{\sigma_{vv}^0 + \sigma_{vh}^0}{2} \right)^{avl} - \left(\frac{\sigma_{vv}^0 + \sigma_{vh}^0}{2} \right)^{ref} \quad (14)$$

ΔH - Scattering entropy variations:

$$\Delta H = H^{avl} - H^{ref} \quad (15)$$

$\Delta\alpha$ Scattering alpha variations

$$\Delta\alpha = \alpha^{avl} - \alpha^{ref} \quad (16)$$

1.4.2 Detection approach

This section describes the development of the proposed approach for snow cover detection. In the MODIS snow cover products, snow cover and non-snow cover were taken as the image objects. The training sample composed of the scattering characteristic variations was developed on the basis of the established image objects. The PCA-SVM object-oriented classification method was adopted for snow cover detection in the study area (Figure 4).

In this study, the training dataset was composed of 200 samples with 124 snow cover samples and 76 non-snow cover samples. The optimum parameters of the SVM were established through a five-fold cross-validation before the whole dataset was classified. For the training dataset, the fine Gaussian kernel was used because of its robustness and potential over other kernels. The classification accuracy of the training samples reached 77.2%. We found the C value of 100 and γ value of 0.06 to work the best.

1.5 Performance evaluation

The snow cover obtained in this study was scored using the following test methods:

$$TS_k = \frac{NA_k}{NA_k + NB_k + NC_k} \times 100\% \quad (17)$$

$$POD_k = \frac{NA_k}{NA_k + NB_k} \times 100\% \quad (18)$$

$$FAR_k = \frac{NC_k}{NA_k + NC_k} \times 100\% \quad (19)$$

$$FB_k = \frac{NA_k + NC_k}{NA_k + NB_k} \times 100\% \quad (20)$$

Eqs. 17–20 represent the threat score (TS), probability of detection (POD), false alarm ratio (FAR), and frequency bias (FB). In these formulas, NA_k is the number of correct classification grids for the snow-covered pixels, NB_k is the number of misclassification grids for the snow-covered pixels, NC_k is the number of false classification grids for the snow-covered pixels, ND_k is the number of grids indicating the non-snow cover in the image and estimation, and k is used to distinguish the

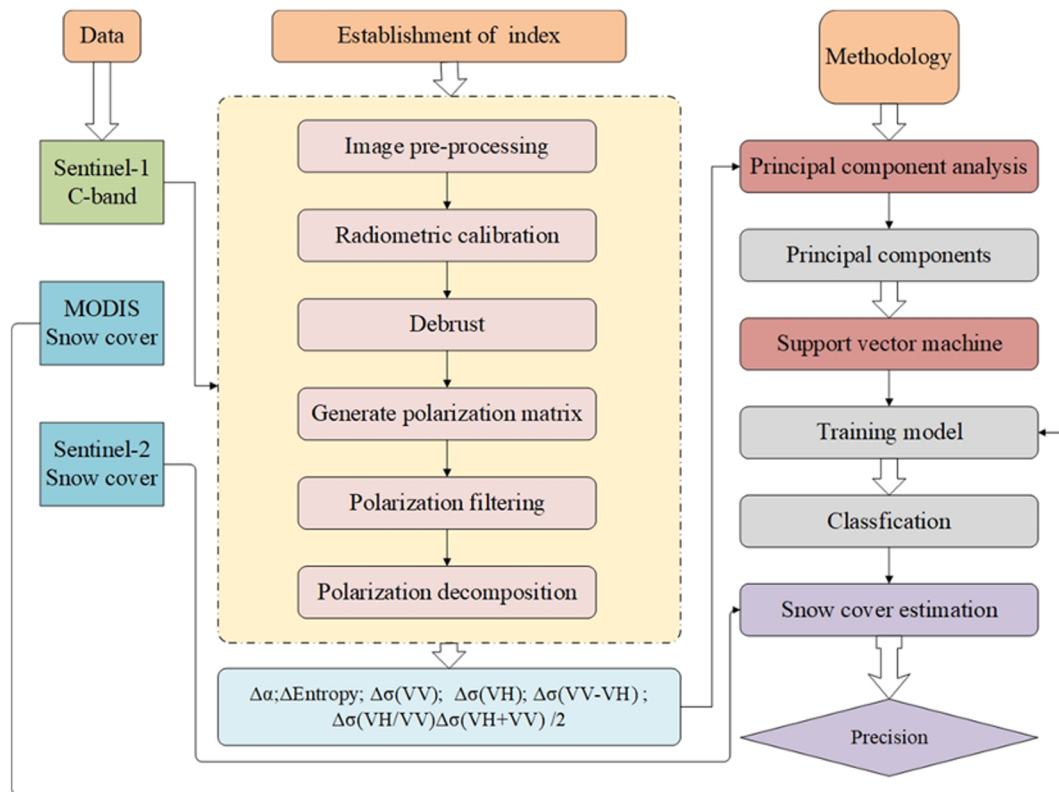


Figure 4 Flowchart of the study. ($\Delta\sigma(VH)$ - VH scattering variations; $\Delta\sigma(VV)$ - VV scattering variations; $\Delta\sigma(VH/VV)$ - Cross-polarized ratio variations; $\Delta\sigma(VV-VH)$ - Polarization difference variations; $\Delta\sigma(VV+VH)/2$ - Polarization mean variations; ΔH - Scattering entropy variations; $\Delta\alpha$ - Scattering alpha variations)

snow cover results of the two snow periods.

2 Results

2.1 Generation of the optimum feature combination from Sentinel-1

The variation of the land surface scattering characteristics is caused by snow cover during the stable period. In this study, “Feb. 10, 2018” of the activity image in Table 1 was selected as the snow cover image in the stable period, and “Nov. 30, 2017” in the reference image was selected as the non-snow cover image. According to the seven analytical indicators ($\Delta\sigma_{vh}$, $\Delta\sigma_{vv}$, $\Delta\sigma_{vh/vv}$, $\Delta\sigma_{vh-vv}$, $\Delta\sigma_{(vh+vv)/2}$, ΔH , and $\Delta\alpha$), these two images were used in the calculation of the variations of snow cover during the stable period. To represent snow cover, PCs were obtained from the snow cover-led scattering characteristic variations using the PCA.

Figure 5 and Table 2 show the graph of the explained variance ratio of the PCs and the cumulative explained variance. According to the

PCA, the selected PCs reflected sufficient information of the original factor, with the cumulative contribution rate of the PCs reaching 85%. As shown in Figure 5, the cumulative contribution rate of PC1 and PC2 was 87.22%. Thus, the first two PCs were analyzed.

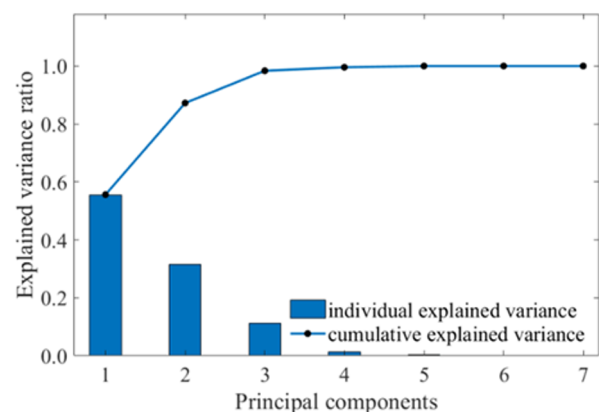


Figure 5 Explained variance ratio of the principal components of 7 selected indicators.

PCA was also applied to rank the features according to their importance. As shown in Table 2, PC1 explained 55.60% of the total variance. In PC1,

a strong positive loading was observed because of the variation of the cross-polarized ratio and the polarization difference between snow cover and non-snow cover. The variations of the scattering alpha between snow cover and non-snow cover showed a negative loading in PC1. PC2 represented 31.62% of the total variance explained. A positive loading was observed because of the variation of the VH backscatter and the polarization mean between snow cover and non-snow cover. It reflected the variations of the scattering randomness and backscatter in the variation of the scattering characteristics in the sample caused by the difference between snow cover and non-snow cover.

2.2 Snow cover mapping from Sentinel-1 and MODIS

According to the detection approach in Section 2.2.2, [Figures 6\(a–b\)](#) illustrate the spatial distribution of snow cover in the study area on January 29, 2018 (a) and February 22, 2018 (b), respectively.

As shown in [Figure 6](#), in terms of the spatial characteristics of the snow pixel distribution, the snow pixels obtained from the PCs of the February 22, 2018 and January 29, 2018 SAR images showed a patchy orderly distribution under the SVM classification method. Although the complex mountain terrain exerted a negative effect on snow detection in the SAR microwave remote sensing data, a certain integrity in the snow cover was distinguishable from the non-snow pixels. As shown in [Figures 6\(a–b\)](#), the snow cover distribution in the study area was mainly concentrated in the high-altitude area of the north and south, consistent with the actual observation.

2.3 Overall results

Snow cover distribution was obtained from Sentinel-2 and served as the true value in the verification of snow cover distribution formed by

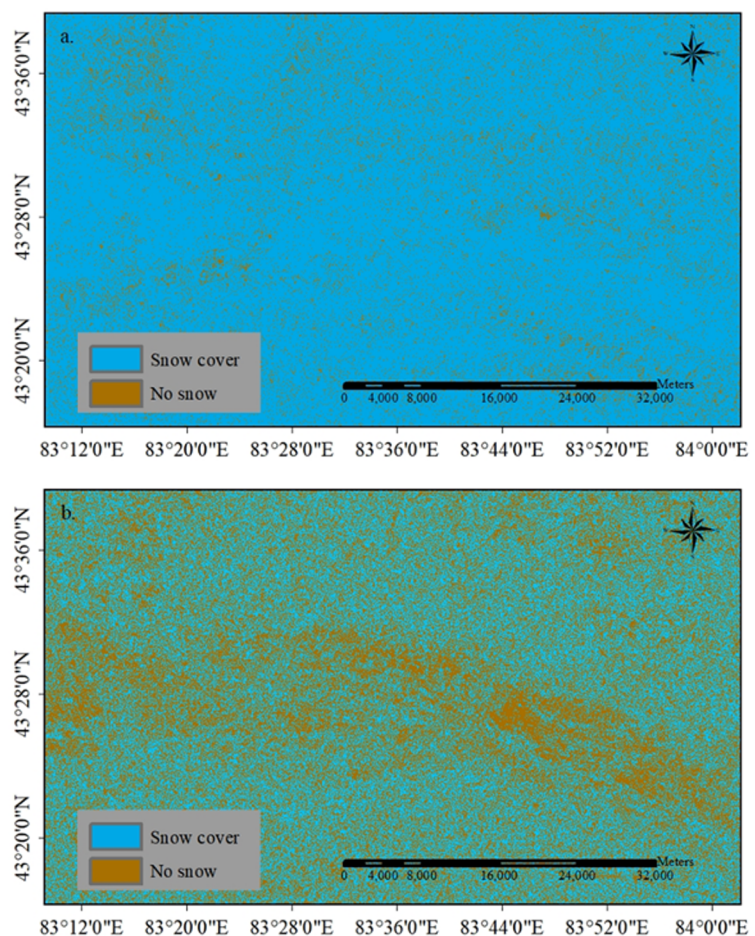


Figure 6 Maps of snow cover in Xinyuan County, western part of the Tianshan Mountains from Support Vector Machine (SVM) algorithm on Jan. 29, 2018 (a), Feb. 22, 2018 (b).

the SAR data. In this section, an accuracy test was conducted using an error matrix based on the training and test area mask with statistical scores, such as frequency bias (FB), hit rate (POD), threat score (TS), and false alarm ratio (FAR), which serve as the accuracy assessments for snow cover. The result is shown in [Table 3](#).

As shown in [Table 3](#), in the February 22, 2018 and January 29, 2018 SAR images obtained by the SVM method, both frequency biases were less than 1. The snow pixels were less in this approach than in the actual situation. The TS, FAR, and POD scores showed some misclassification and false classification in the snow pixels. Moreover, the precision of snow cover pixels obtained in the January 29, 2018 period was better than that obtained in the February 22, 2018 period. Some differences were found in each statistical indicator, indicating that the detection of snow pixel by the C-

band from the Sentinel-1 SAR could be affected by the variations in snow characteristics and ground surface. Overall, the SVM-based object classification yielded pertinent extraction results of snow pixels in a stable period.

Table 2 Principal components of snow avalanche characteristics

Indicators	PC1	PC2
ΔH	-0.4242	0.0541
$\Delta \alpha$	0.4263	-0.0112
$\Delta \sigma_{VV}$	0.3645	0.2918
$\Delta \sigma_{VH}$	-0.2034	0.4019
$\Delta \sigma_{VH/VV}$	0.4793	-0.0284
$\Delta \sigma_{VV-VH}$	0.4793	-0.0284
$\Delta \sigma_{(VV+VH)/2}$	0.0676	0.4477
Eigen values	3.89	2.21
% of variance	55.60	31.62
Cumulative %	55.60	87.22

Notes: Values in bold represent significant contribution by a particular variable. ($\Delta \sigma(VH)$ - VH scattering variations; $\Delta \sigma(VV)$ - VV scattering variations; $\Delta \sigma_{VH/VV}$ - Cross-polarized ratio variations; $\Delta \sigma_{VV-VH}$ - Polarization difference variations; $\Delta \sigma_{(VV+VH)/2}$ - Polarization mean variations; ΔH - Scattering entropy variations; $\Delta \alpha$ - Scattering alpha variations)

Table 3 Statistics assessments for the snow cover classification. TS = scores of the threat score; FAR = false alarm ratio; POD = probability of detection; FB = frequency bias.

Time point	TS	FAR	POD	FB
Jan. 29, 2018	86.84	4.01	9.01	93.86
Feb 22, 2018	56.40	3.62	57.62	59.78

3 Discussion

3.1 Comparison of the different schemes for snow cover estimation

Snow cover detection is an important research topic in snow remote sensing. Currently, visible/short-wave infrared and passive microwave bands are used to obtain the snow cover distribution (Czyzowska-Wisniewski et al. 2015; Armstrong and Brodzik 2001; Foster et al. 2011). Visible/short-wave infrared is based on the high reflectivity of snow cover in the visible band and the low reflectivity of snow cover in the short-wave infrared band. The passive microwave band is based on the attenuation of snow cover in brightness. Generally, the visible/short-wave infrared band is suitable for acquiring snow cover in medium-scale target areas because of its high

spatial resolution, but it is vulnerable to influences from clouds. The spatial resolution of passive microwave remote sensing data is coarse, thus making it only suitable for large-scale snow cover estimation. In recent years, active microwave remote sensing has also been used for snow cover detection (Pivot 2012; Tedesco and Miller 2017; Shi 2008; Nandan et al. 2016; Rott 1994; Tsai et al. 2019). Although this method overcomes the block from cloud and fog, snow cover distribution is underestimated.

To fully take advantage of the Sentinel-1 SAR images, this study proposed the PCA-SVM method to obtain the snow cover distribution as it amplifies the advantages of a reduced data dimension, a small sample size, and scale transformation. The SVM classifier used in the study had a high accuracy (AUC = 0.772). If other classifiers, such as the regression-tree classifier (AUC = 0.700) and the artificial neural network classifier (AUC = 0.726) were used, the accuracy would be lower. Generally, despite the issues of omission and misclassification, compared with the snow cover traditionally obtained by a threshold algorithm from optical remote sensing images, the PCA-SVM method can integrate different images with diverse temporal and spatial resolutions, which compensate for the limitations of a single image with the two advantages.

3.2 Error source in snow pixel detection

Compared with that of homogeneous undisturbed fresh snow, the density of the snowpack in a stable period increases because of compaction and repeated freezing and thawing, thus forming a deep frost in the snow layer. Previous studies showed that the active microwave C-band could penetrate the snowpack up to 20 m, but the inhomogeneity and the liquid water content of the snowpack affect its dielectric properties, which change the scattering characteristics of the snowpack (Dedieu et al. 2014; Bilal et al. 2019; Bukovčukovl. 2014; Bilal et al. 2019; undisturbed fresh snow.). In this study, the snow depth in the observation point of the study area was 10.00–40.00 cm, the deep frost layer was 4.00–10.00 cm, the water content in the snowpack was 0.11%–0.34%, and the density was 0.13–0.24 g/cm³. Compared with those of non-snow cover, the snow

characteristics rendered the attenuation of the backscatter in the VH/VV mode. The mean alpha angle (α) increased slightly, and entropy (H) decreased to a certain degree unlike in non-snow cover. The data in these observation points were in accordance with the above theoretical foundation (Figure 7(a-b)). However, the variations of the polarization parameters were not sufficient to differentiate snow from non-snow cover (Figure 7(c-d)). As a result, the phenomena of “the same object with different spectra” and “the same spectra with different objects” were inevitable, thus leading to misclassification and false detection in the snow pixel classification.

3.3 Limitations and future work

The SAR data in this study were not used in the L and X bands, which perform better in interacting with the snowpack. Furthermore, instead of obtaining the snow cover distribution using the commonly employed full-polarization mode of SAR, this study used the C-band dual-polarization from the Sentinel-1 SAR data. The

VV/VH dual polarization mode was used exclusively, and thus the HH and HV polarization modes were not included in Sentinel-1. the H/α plane only reflects double bounce scattering (Figure 8).

Although SVM was used as the object-oriented classification method for snow cover detection, its model accuracy was limited by the quality of the input training samples. In addition, the classification algorithm limited by the model could not fully establish the nonlinear relationship between the feature parameters and the snow cover.

In subsequent studies, further consideration should be given to other methods for supervised classification. The method of supervised classification for snow cover estimation could be improved further because of the available access to remote sensing data and the extensible variables of snow sensitivity.

4 Conclusion

The method of obtaining the snow cover spatial distribution in this study overcame the

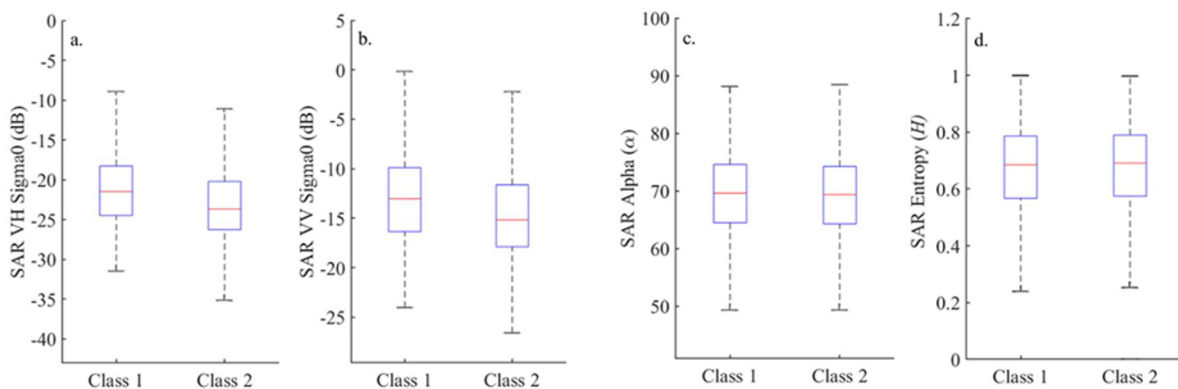


Figure 7 Box and whisker plot for the geometric feature of the VV/VH backscatter (a-b), alpha and entropy (c-d) variations caused by snow cover in stable period.

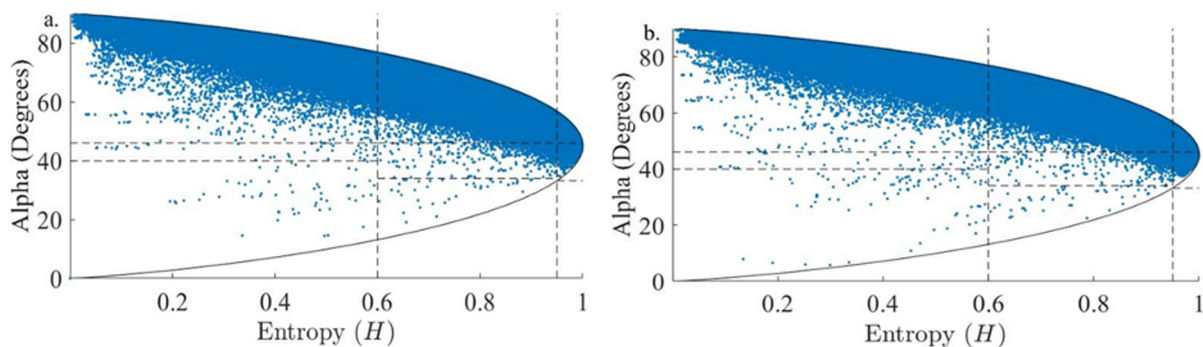


Figure 8 Distribution of entropy (H)-alpha (α) plot from Sentinel-1 SAR on Nov. 30, 2018 (a), Feb 10, 2018 (b).

limitations of snow cover detection using a single remote sensing data source and a statistical algorithm of the past. On the one hand, it reduced the effect imposed by cloud, fog, and terrain on the snow pixel detection of a complex terrain. On the other hand, it exploited the advantages of SVM in a small sample size and better integrated the different spatial resolution images.

This study examined the capacity of Sentinel-1 SAR data for snow cover detection. Seven different indicators were established from the variation of the scattering characteristics caused by snow cover along with MODIS snow cover products. The approach for snow cover estimation using machine learning methods, including PCA and SVM, optimized the integration of images with the different spatial resolutions. The fine extraction of the 20 m spatial resolution of snow cover distribution in the study area was achieved. The snow cover estimation approach proposed in this study was found to be applicable in the study area during a stable period. Accurately distinguishing the variations of the scattering characteristics

caused by snow cover or the ground surface is difficult. Thus, to a certain extent, the proposed method resulted in false classification and misclassification in the snow cover detection because of the insensitivity of the C-band to the changes in the land surface caused by snow cover.

Acknowledgments

This study was supported by the Open Project of Key Laboratory, Xinjiang Uygur Autonomous Region (No. 2019D04003), the National Natural Science Foundation of China (NSFC Grant U1703241, 41901087), the West Light Foundation of the Chinese Academy of Sciences (No. 2018-XBQN-Z-B-012), the Key International cooperation project of Chinese Academy of Sciences (No: 121311KYSB20160005), and the CAS Instrumental development project of Automatic Meteorological Observation System with Multifunctional Modularization (No: Y634241001).

References

- Armstrong RL, Brodzik MJ (2001) Recent Northern Hemisphere snow extent: A comparison of data derived from visible and microwave satellite sensors. *Geophysical Research Letters* 28(19): 3673–3676.
<https://doi.org/10.1029/2000GL012556>
- Bernier M, Fortin JP (1998) The potential of time series of C-Band SAR data to monitor dry and shallow snow cover. *IEEE Transactions on Geoscience and Remote Sensing* 36(1): 226–243. <https://doi.org/10.1109/36.655332>
- Berman EE, Bolton DK, Coops NC, et al. (2018) Daily estimates of Landsat fractional snow cover driven by MODIS and dynamic time-warping. *Remote Sensing of Environment* 216: 635–646. <https://doi.org/10.1016/j.rse.2018.07.029>
- Buković P, Ryzhkov A, Zrnić D, et al. (2018) Polarimetric radar relations for quantification of snow based on disdrometer data. *Journal of Applied Meteorology and Climatology* 57(1): 103–120. <https://doi.org/10.1175/JAMC-D-17-0090.1>
- Bilal H, Chamhuri S, Mokhtar M, et al. (2019) Recent snow cover variation in the Upper Indus Basin of Gilgit Baltistan, Hindukush Karakoram Himalaya. *Journal of Mountain Science* 16(2): 296–308.
<https://doi.org/10.1007/s11629-018-5201-3>
- Bangira T, Alfieri SM, Menenti M, et al. (2019) Comparing Thresholding with Machine Learning Classifiers for Mapping Complex Water. *Remote Sensing* 11(11): 1351.
<https://doi.org/10.3390/rs11111351>
- Chen Y, Tan K, Yan SY, et al. (2019) Monitoring Land Surface Displacement over Xuzhou (China) in 2015–2018 through PCA-Based Correction Applied to SAR Interferometry. *Remote Sensing* 231: 111235.
<https://doi.org/10.3390/rs11121494>
- Cloude SR (2007) The Dual Polarisation entropy/alpha decomposition: A PALSAR case study. In *POLinSAR`07: the 3rd International Workshop on Science and Applications of SAR Polarimetry and Polarimetric Interferometry*, Frascati, Italy, January 22–26.
<http://adsabs.harvard.edu/abs/2007ESASP.644E...2C>
- Czyzowska-Wisniewski EH, Van Leeuwen WJD, Hirschboeck KK, et al. (2015) Fractional snow cover estimation in complex alpine-forested environments using an artificial neural network. *Remote Sensing of Environment* 156: 403–417.
<https://doi.org/10.1016/j.rse.2014.09.026>
- Dedieu JP, Besic N, Vasile G, et al. (2014) Dry snow analysis in alpine regions using RADARSAT-2 full polarimetry data. Comparison with in situ measurements. *International Geoscience and Remote Sensing Symposium (IGARSS)* 3658–3661. <https://doi.org/10.1109/IGARSS.2014.6947276>
- Dietz AJ, Kuenzer C, Gessner U, et al. (2012) Remote sensing of snow - a review of available methods. *International Journal of Remote Sensing* 33(13): 4094–4134.
<https://doi.org/10.1080/01431161.2011.640964>
- Forman BA, Reichle RH (2015) Using a Support Vector Machine and a Land Surface Model to Estimate Large-Scale Passive Microwave Brightness Temperatures over Snow-Covered Land in North America. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 8(9): 4431–4441. <https://doi.org/10.1109/JSTARS.2014.2325780>
- Foster JL, Hall DK, Eylander JB, et al. (2011) A blended global snow product using visible, passive microwave and scatterometer satellite data. *International Journal of Remote Sensing* 32(5): 1371–1395.
<https://doi.org/10.1080/01431160903548013>
- Gafurov A, Bárdossy A (2009) Cloud removal methodology from MODIS snow cover product. *Hydrology and Earth System Sciences* 13(7): 1361–1373.
<https://doi.org/10.5194/hess-13-1361-2009>

- Gao Y, Xie H, Lu N, et al. (2010) Toward advanced daily cloud-free snow cover and snow water equivalent products from Terra-Aqua MODIS and Aqua AMSR-E measurements. *Journal of Hydrology* 385(1-4): 23-35. <https://doi.org/10.1016/j.jhydrol.2010.01.022>
- Gascoin S, Grizonnet M, Bouchet M, et al. (2019) Theia Snow collection: High-resolution operational snow cover maps from Sentinel-2 and Landsat-8 data. *Earth System Science Data* 11(2): 493-514. <https://doi.org/10.5194/essd-11-493-2019>
- Hall DK, Riggs GA (2007) Accuracy assessment of the MODIS snow products. *Hydrological Processes* 21: 1534-1547. <https://doi.org/10.1002/hyp.6715>
- Hao JS, Huang FR, Liu Y, et al. (2018) Avalanche activity and characteristics of its triggering factors in the western Tianshan Mountains, China. *Journal of Mountain Science* 15(7): 1397-1411. <https://doi.org/10.1007/s11629-018-4941-2>
- Hou J, Huang C, Zhang Y, et al. (2019) Gap-Filling of MODIS Fractional Snow Cover Products via Non-Local Spatio-Temporal Filtering Based on Machine Learning Techniques. *Remote Sensing* 11(1): 90. <https://doi.org/10.3390/rs11010090>
- Huang Y, Liu H, Yu B, et al. (2018) Improving MODIS snow products with a HMRP-based spatio-temporal modeling technique in the Upper Rio Grande Basin. *Remote Sensing of Environment* 204: 568-582. <https://doi.org/10.1016/j.rse.2017.10.001>
- He G, Feng X, Xiao P, et al. (2017) Dry and wet snow cover mapping in mountain areas using SAR and optical remote sensing data. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 10(6): 2575-2588. <https://doi.org/10.1109/JSTARS.2017.2673409>
- Kostadinov TS, Schumer R, Hausner M, et al. (2019) Watershed-scale mapping of fractional snow cover under conifer forest canopy using lidar. *Remote Sensing of Environment* 222: 34-49. <https://doi.org/10.1016/j.rse.2018.11.037>
- Kuter S, Akyurek Z, Weber GW (2018) Retrieval of fractional snow covered area from MODIS data by multivariate adaptive regression splines. *Remote Sensing of Environment* 205: 236-252. <https://doi.org/10.1016/j.rse.2017.11.021>
- Liu Y, Li LH, Yang JM, et al. (2017) Estimating snow depth using multi-source data fusion based on the D-InSAR method and 3DVAR fusion algorithm. *Remote Sensing* 9(11): 1195. <https://doi.org/10.3390/rs9111195>
- Li Y, Chen Y, Li, Z (2019) Developing Daily Cloud-Free Snow Composite Products From MODIS and IMS for the Tianshan Mountains. *Earth and Space Science* 6(2): 266-275. <https://doi.org/10.1029/2018EA000460>
- Lima AR, Cannon AJ, Hsieh WW (2015) Nonlinear regression in environmental sciences using extreme learning machines: A comparative evaluation. *Environmental Modelling and Software* 73: 175-188. <https://doi.org/10.1016/j.envsoft.2015.08.002>
- Lu H, Wei WS, Liu MZ, et al. (2016) Variations in seasonal snow surface energy exchange during a snowmelt period: an example from the Tianshan Mountains, China. *Meteorological Applications* 23(1): 14-25. <https://doi.org/10.1002/met.1511>
- Liu JF, Chen RS, Wang G (2015) Snowline and snow cover monitoring at high spatial resolution in a mountainous river basin based on a time-lapse camera at a daily scale. *Journal of Mountain Science* 12(1): 60-69. <https://doi.org/10.1007/s11629-013-2842-y>
- Li X, Fu W, Shen H, et al. (2017) Monitoring snow cover variability (2000-2014) in the Hengduan Mountains based on cloud-removed MODIS products with an adaptive spatio-temporal weighted method. *Journal of hydrology* 551: 314-327. <https://doi.org/10.1016/j.jhydrol.2017.05.049>
- Liu Y, Li LH, Chen X, et al. (2018) Spatial distribution of snow depth based on geographically weighted regression kriging in the Bayanbulak Basin of the Tianshan Mountains, China. *Journal of Mountain Science* 15(1): 33-45. <https://doi.org/10.1007/s11629-017-4564-z>
- Li Q, Yang T, Zhang FY, et al. (2019) Snow depth reconstruction over last century: Trend and distribution in the Tianshan Mountains, China. *Global and Planetary Change* 173: 73-82. <https://doi.org/10.1016/j.gloplacha.2018.12.008>
- Li X, Jing Y, Shen H, et al. (2019) The recent developments in cloud removal approaches of MODIS snow cover product. *Hydrology and Earth System Sciences* 23(5): 2401-2416. <https://doi.org/10.5194/hess-23-2401-2019>
- Li X, Jing Y, Shen H, et al. (2019) The recent developments in spatio-temporally continuous snow cover product generation. *Hydrology and Earth System Sciences Discussions* 1-28. <https://doi.org/10.5194/hess-2018-633>
- Marshall S, Roads JO, Glatzmaier G (1994) Snow hydrology in a general circulation model. *Journal of Climate* 7(8): 1251-1269. [https://doi.org/10.1175/1520-0442\(1994\)007<1251:SHIAGC>2.0.CO;2](https://doi.org/10.1175/1520-0442(1994)007<1251:SHIAGC>2.0.CO;2)
- Mityók ZK, Bolton DK, Coops NC, et al. (2018) Snow cover mapped daily at 30 meters resolution using a fusion of multi-temporal MODIS NDSI data and Landsat surface reflectance. *Canadian Journal of Remote Sensing* 1-22. <https://doi.org/10.1080/07038992.2018.1538775>
- Masson T, Dumont M, Mura MD, et al. (2018) An assessment of existing methodologies to Retrieve Snow Cover Fraction from MODIS Data. *Remote Sensing* 10(4): 619. <https://doi.org/10.3390/rs10040619>
- Nijhawan R, Das, J, Raman, B (2019) A hybrid of deep learning and hand-crafted features based approach for snow cover mapping. *International Journal of Remote Sensing* 40(2): 759-773. <https://doi.org/10.1080/01431161.2018.1519277>
- Nandan V, Geldsetzer T, Islam T, et al. (2016) Ku-, X- and C-band measured and modeled microwave backscatter from a highly saline snow cover on first-year sea ice. *Remote Sensing of Environment* 187: 62-75. <https://doi.org/10.1016/j.rse.2016.10.004>
- Pivot FC (2012) C-band sar imagery for snow-cover monitoring at treeline, churchill, manitoba, Canada. *Remote Sensing* 4(7): 2133-2155. <https://doi.org/10.3390/rs4072133>
- Parajka J, Blöschl G (2006) Validation of MODIS snow cover images over Austria. *Hydrology and Earth System Sciences* 3(4): 1569-1601. <https://doi.org/10.5194/hess-10-679-2006>
- Pearson K (1901) On lines and planes of closets fit to systems of points in the space. *Philosophical Magazine* 6: 559-572. <https://doi.org/10.1080/14786440109462720>
- Qiao D, Wang N (2019) Relationship between Winter Snow Cover Dynamics, Climate and Spring Grassland Vegetation Phenology in Inner Mongolia, China. *ISPRS International Journal of Geo-Information* 8(1): 42. <https://doi.org/10.3390/ijgi8010042>
- Xie Q, Meng Q, Zhang L, et al. (2018) Combining of the H/A/Alpha and Freeman-Durden Polarization Decomposition Methods for Soil Moisture Retrieval from Full-Polarization Radarsat-2 Data. *Advances in Meteorology* 1-17. <https://doi.org/10.1155/2018/9436438>
- Rott H (1994) Snow Mapping in Alpine Regions with Synthetic Aperture Radar. *IEEE Transactions on Geoscience and Remote Sensing* 5(1): 129. <https://doi.org/10.1109/36.285197>
- Rondeau-Genesse G, Trudel M, Leconte R (2016) Monitoring snow wetness in an Alpine Basin using combined C-band SAR and MODIS data. *Remote Sensing of Environment* 183: 304-317. <https://doi.org/10.1016/j.rse.2016.06.003>
- Şorman AA, Uysal G, Şensoy A (2019) Probabilistic snow cover and ensemble streamflow estimations in the upper euphrates basin. *Journal of Hydrology and Hydromechanics* 67(1): 82-92. <https://doi.org/10.2478/johh-2018-0025>
- Snapir B, Momblanch A, Jain SK, et al. (2019) A method for monthly mapping of wet and dry snow using Sentinel-1 and MODIS: Application to a Himalayan river basin. *International Journal of Applied Earth Observation and Geoinformation* 74: 222-230. <https://doi.org/10.1016/j.jag.2018.09.011>
- Shi JC, Dozier J (1997) Mapping seasonal snow with SIR-C/X-SAR in mountainous areas. *Remote Sensing of Environment* 59(2): 294-307.

- [https://doi.org/10.1016/S0034-4257\(96\)00146-0](https://doi.org/10.1016/S0034-4257(96)00146-0)
- Salzano R, Salvatori R, Valt M, et al. (2019) Automated Classification of Terrestrial Images: The Contribution to the Remote Sensing of Snow Cover. *Geosciences* 9(2): 97. <https://doi.org/10.3390/geosciences9020097>
- Shi JC (2008) Active Microwave Remote Sensing Systems and Applications to Snow Monitoring. In *Advances in Land Remote Sensing*. https://doi.org/10.1007/978-1-4020-6450-0_3
- Shi JC, Dozier J (1995) Inferring Snow Wetness Using C-Band Data from SIR-C's Polarimetric Synthetic Aperture Radar. *IEEE Transactions on Geoscience and Remote Sensing* 33(4): 905-914. <https://doi.org/10.1109/36.406676>
- Shi JC, Dozier J (2000) Estimation of snow water equivalence using SIR-C/X-SAR. I. Inferring snow density and subsurface properties. *IEEE Transactions on Geoscience and Remote Sensing* 38(6): 2465-2474. <https://doi.org/10.1117/12.319499>
- Shui T, Liu J, Xiao Y, et al. (2019) Effects of snow cover on urban surface energy exchange: Observations in Harbin, China during the winter season. *International Journal of Climatology* 39(3): 1230-1242. <https://doi.org/10.1002/joc.5873>
- Tran H, Nguyen P, Ombadi M, et al. (2019) A cloud-free modis snow cover dataset for the contiguous United States from 2000 to 2017. *Scientific Data* 6: 180300. <https://doi.org/10.1038/sdata.2018.300>
- Tsai YLS, Dietz A, Oppelt N, et al. (2019) Remote Sensing of Snow Cover Using Spaceborne SAR: A Review. *Remote Sensing* 11(12): 1456. <https://doi.org/10.3390/rs11121456>
- Tedesco M, Miller J (2007) Northern hemisphere snow-covered area mapping: Optical versus active and passive microwave data. *IEEE Geoscience and Remote Sensing Letters* 4(2): 221-225. <https://doi.org/10.1109/LGRS.2006.888842>
- Tamocai C, Canadell JG, Schuur EAG, et al. (2009) Soil organic carbon pools in the northern circumpolar permafrost region. *Global Biogeochemical Cycles* 23(2): 2607-2617. <https://doi.org/10.1029/2008GB003327>
- Tsai YLS, Dietz A, Oppelt N, et al. (2019) Wet and dry snow detection using sentinel-1 SAR data for mountainous areas with a machine learning technique. *Remote Sensing* 11(8): 895. <https://doi.org/10.3390/rs11080988>
- Vignols RM, Marshall GJ, Rees WG, et al. (2019) Assessing snow cover changes in the Kola Peninsula, Arctic Russia, using a synthesis of MODIS snow products and station observations. *The Cryosphere Discussions* 1-33. <https://doi.org/10.5194/tc-2019-9>
- Wu S, Zhang X, Du J, et al. (2019) The vertical influence of temperature and precipitation on snow cover variability in the Central Tianshan Mountains, Northwest China. *Hydrological Processes* 1-12. <https://doi.org/10.1002/hyp.13431>
- Wu J, Wu Z (2019) Inter-decadal change of the spring North Atlantic Oscillation impact on the summer Pamir-Tianshan snow cover. *International Journal of Climatology* 39(2): 629-642. <https://doi.org/10.1002/joc.5831>
- Wei Y, Wang S, Fang Y, et al. (2017) Integrated assessment on the vulnerability of animal husbandry to snow disasters under climate change in the Qinghai-Tibetan Plateau. *Global and Planetary Change* 157: 139-152. <https://doi.org/10.1016/j.gloplacha.2017.08.017>
- Whyte A, Ferentinos KP, Petropoulos GP (2018) A new synergistic approach for monitoring wetlands using Sentinels -1 and 2 data with object-based machine learning algorithms. *Environmental Modelling and Software* 104: 40-54. <https://doi.org/10.1016/j.envsoft.2018.01.023>
- Xiao X, Zhang T, Zhong X, et al. (2018) Support vector regression snow-depth retrieval algorithm using passive microwave remote sensing data. *Remote Sensing of Environment* 210: 48-64. <https://doi.org/10.1016/j.rse.2018.03.008>
- Xu G, Zhu X, Fu D, et al. (2017) Automatic land cover classification of geo-tagged field photos by deep learning. *Environmental Modelling and Software* 97: 127-134. <https://doi.org/10.1016/j.envsoft.2017.02.004>
- Zhang T (2005) Influence of the seasonal snow cover on the ground thermal regime: An overview. *Reviews of Geophysics* 43(4): 589-590. <https://doi.org/10.1029/2004RG000157>
- Zimov SA, Schuur EAG, Chapin FS (2006) Permafrost and the global carbon budget. *Science* 312(5780): 1612-1613. <https://doi.org/10.1126/science.1128908>
- Zhang H, Zhang F, Zhang G, et al. (2019) Ground-based evaluation of MODIS snow cover product V6 across China: Implications for the selection of NDSI threshold. *Science of the Total Environment* 651: 2712-2726. <https://doi.org/10.1016/j.scitotenv.2018.10.128>